SADIA ANJUM MIM

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RESEARCH INTERESTS

Computational Fluid Dynamics (CFD); Machine Learning; Microfluidics; Design Optimization; MEMS and Microfabrication

EDUCATION

Washington State University Vancouver Aug. 2025 – May 2027 (expected)
M.S. in Mechanical Engineering GPA: 3.93/4.00 *Vancouver, WA, USA*

Bangladesh University of Engineering and Technology (BUET) Apr. 2019 – Jul. 2024
B.Sc. in Mechanical Engineering GPA: 3.57/4.00 *Dhaka, Bangladesh*
(Dean's List and Merit Scholarship for 2 Semesters)

PUBLICATIONS

Journal Articles

[J1] **S. A. Mim**, M. R. Uddin, and X. Chen, "Bayesian optimization of deterministic lateral displacement devices for circulating tumor cell separation," *Biomedical Microdevices*, vol. 28, art. no. 72, 2026. doi: 10.1007/s10544-026-00856-4

[J2] T. Sarowar, **S. A. Mim**, and X. Chen, "Data-driven design of deterministic lateral displacement microfluidic devices for CTC separation," *Biotechnology Progress*, vol. 42, no. 4, e88515, 2026. doi: 10.1002/btpr.88515

Conference Papers

[C1] **S. A. Mim**, M. R. Uddin, and X. Chen, "Optimizing microfluidic CTC separation using Bayesian methods," in *Proc. ASME 2026 Fluids Engineering Division Summer Meeting (FEDSM2026)*, Bellevue, WA, USA, Jul. 26–29, 2026, Paper No. FEDSM2026-184031.

Manuscripts Under Review

[R1] P. Maggu, **S. A. Mim**, M. R. Uddin, and X. Chen, "Surrogate-level versus acquisition-level transfer learning for DLD optimization," submitted to *IEEE International Conference on Big Data (BigData 2026)*.

[R2] V. Varikooty, **S. A. Mim**, M. R. Uddin, and X. Chen, "Adaptive source trust in transfer Bayesian optimization: Gated RGPE versus TST-R for DLD design," submitted to *IEEE International Conference on Big Data (BigData 2026)*.

[R3] R. S. Alif, **S. A. Mim**, M. K. Alom, and M. Foysal, "A robust hybrid deep ensemble framework for MRI-based brain tumor classification using power-law preprocessing," *Computer Methods and Programs in Biomedicine Update*, revised manuscript under review.

RESEARCH EXPERIENCE

Graduate Research, Computer-Aided Engineering Lab Aug. 2025 – Present
Advisor: Dr. Xiaolin (Linda) Chen, Associate Professor *Dept. of ME, WSU*
Research: *Transfer-enhanced Bayesian optimization for microfluidic CTC separation*

- Developed a simulation-driven Bayesian optimization framework (Python + COMSOL) for DLD design, reducing simulations by 67.2% and time-to-solution by 80% vs. ES, outperforming GA, DE, and PSO.
- Developing multi-objective and transfer-learning methods to balance CTC separation against pressure drop and to reuse knowledge across related DLD design tasks

Undergraduate Thesis Research, Macro to Micro Scale Fluids Engineering Lab Jul. 2023 – Jun. 2024
Advisor: Dr. A. B. M. Toufique Hasan, Professor *Dept. of ME, BUET*
Thesis: *Study of non-Newtonian fluid flow past a circular cylinder at low Reynolds number*

- Investigated Newtonian and shear-thinning power-law fluid flow ($n = 0.2$–1.0) past a circular cylinder at $Re = 0.1$–40 using ANSYS Fluent, focusing on drag reduction and confinement effects.

Teaching and Mentoring

Graduate Teaching Assistant Aug. 2025 – Present
School of Engineering and Computer Science, Washington State University Vancouver *Vancouver, WA*

- MECH 251 Numerical Computing for Engineers; MECH 314 Machine Design I; MECH 414 Machine Design II; MECH 485 Computer-Aided Engineering. Office hours, grading, and one-on-one technical guidance.
- Research mentor to two undergraduate students in the NSF Research Experiences for Undergraduates (REU) program, Summer 2026.

Selected Projects

- **ML-based classification of volcanic ash particle collisions** (ongoing): building a particle-motion simulator and labeled dataset to classify rebound, aggregation, and disaggregation events in high-speed imagery.
- **NLP text summarizer** (2024–2025, GitHub): fine-tuned Google's PEGASUS model on the SAMSum dataset; evaluated with ROUGE metrics.
- **Bio-inspired RC robotic fish** (2022, ResearchGate): designed and built an Arduino-controlled, Bluetooth-operated robotic fish mimicking natural fish motion.
- **Finned cross-flow radiator** (2023): designed a double-pass copper-tube radiator with sine-wave aluminum fins; fabricated components and conducted pressure and leak tests.
- **Graphene hybrid battery and torque modulation in EVs** (ResearchGate): technical case presentation on EV batteries, torque control, and thermal management.

Honors and Awards

- Dean's List, BUET (2019–2020); University Merit Scholarship, BUET (2021)
- 3rd Place, Inter-Varsity Engineering Case Competition

Relevant Graduate Coursework

- MEMS Engineering
- Lab-on-a-Chip
- Sensors
- Robotics
- Machine Learning and AI for Engineers
- CFD and Heat Transfer

Technical Skills

Simulation & CFD:	COMSOL Multiphysics, ANSYS, OpenFOAM
Optimization & ML:	Bayesian optimization, Gaussian-process surrogates, Uncertainty Quantification, Transfer learning, PyTorch (CNNs, NLP), Image Processing
Programming:	Python, MATLAB, C, LaTeX
Microfabrication:	Photolithography, spin coating, wet chemical etching, RF magnetron sputtering, thin films
CAD & Design:	SolidWorks, AutoCAD, HTRI, 3E Plus
Hardware & Shop:	Arduino, Raspberry Pi, PLC; welding, milling, lathe, drilling, grinding

Additional Experience and Activities

- Industrial Trainee, Khulna Shipyard Ltd., Bangladesh (Oct.–Nov. 2023): ship production and safety practices
- Member, IMechE BUET Student Chapter, BUET Automobile Club, Debating Club, and Chess Club (prizes), 2019–2023

References

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